

Numerical study on the performance of a floating circular cross section U-tube type wave energy extractor unit in the ocean environment

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ABSTRACT

A wave energy converter (WEC) unit based on a floating U-tube liquid column has been investigated in this paper. The whole system resembles to a tuned liquid column damper supported by two buoys. It is having six rigid body degrees of freedom (DOFs) like any floating object, and additionally the movement of the water column is assumed to be coupled with surge, heave, and pitch of the floating U-tube. A coupled analysis approach has been adopted to solve the problem. The mathematical model of the motion of the liquid column has been obtained through Lagrangian mechanics. Simpler problems related to the liquid column dampers with smaller number of DOFs studied for the vibration suppression of buildings, bridges, etc., whose solutions are well established in the literature, have been used to validate the mathematical model. The power take off system has been modelled as a linear damper to reduce the complexity in the current work. The preliminary dimensions of the current unit have been chosen after a few random trials focussing to generate around 50 kW power from regular waves of around 1–2 m height. Thereafter the sensitivity of the generated power with respect to the basic structural parameters defining the geometry has been studied. The survival performance during an extreme event has also been investigated. The presented results prove the technical feasibility of deploying such a unit for generating the chosen range of renewable power from the ocean waves.

1. Introduction

Electricity is a basic necessity of human civilization. Since the last century, it is primarily being produced by burning fossil fuels, which is directly linked with global warming, climate change, disruption of ecosystems, unpredictable weather patterns, respiratory and cardiovascular diseases, etc. [1]. Therefore, there is a pressing need to shift our dependency from fossil fuels to cleaner renewable energy.

Almost 70 % of earth's crust is covered by the oceans, and abundant amount of energy is available in the form of waves. Compared to various distributed form of renewable energy resources, the energy density in ocean waves is significantly higher [2], ranging from 20 to 70 kW per meter of crest length at 30°–60° latitudes [3]. However, compared to the other renewable energy resources (e.g., solar, wind), the wave energy sector is least developed. This is primarily due to the large capital and operational expenditures involved with the wave energy converters (WECs) [4].

It is however promising to note that the levelized cost of energy for WECs like the oscillating water column (OWC) and the point absorber system have been found to be comparable with that of the wind energy

in UK [5] or even with the solar energy in the Egyptian desert [6]. With continued research, it may be expected that in the near future, wave energy will become more competitive than the other forms of renewable energy. The global wave energy market has been projected to reach 150 million USD by 2029 [7]. More than a thousand proposals for utilizing wave energy have been submitted for patents [8], and these exist at different technology readiness levels (TRLs) as of today. While most of these have been used for research and development, a few have been connected to the power grids, revealing their technical feasibility and the necessity for further research.

Based on the working principles, WECs can be classified as [9]: Pressure Differential, Floating Structure, Overtopping Device, and Impact Device. A brief survey on these devices is presented in the following paragraphs.

The Pressure Differential principle has been used in the oscillating water column (OWC) type devices. As shown in Fig. 1a, it consists of a semi-submerged hollow chamber with an air pocket and a water column. As the air-water interface oscillates due to the incoming waves, the air is forced to move in or out of the air pocket, which is used to drive a bidirectional air turbine. Such a turbine connected to a generator is commonly used as the power take-off (PTO) system in an OWC. This

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Nomenclature			
A	Cross sectional area of tube (m^2)	R_1, R_2	Distance of centre of gravity of water column in left vertical tube and right vertical tube respectively from the COG of complete system (m)
$A_{\infty ij}$	Matrix of added mass (kg)	R_A	Ratio of area of vertical tube to area of horizontal tube
A_b	Total water plane area (m^2)	R_L	Ratio of length of vertical water column to that of horizontal water column
A_H	Area of horizontal tube (m^2)	R_M	Ratio of mass of water column to mass of U-tube and buoys
Amp_{max}	Maximum amplitude	s	Second
Amp_{RMS}	Root mean square amplitude	t	Variable for time
A_V	Area of vertical tube (m^2)	T	Wave period (s)
B_{ij}	Retardation function (Ns/m)	T_h	Time period in heave (s)
C	Damping coefficient (Ns/m)	T_p	Time period in pitch (s)
C_A, C_D	Coefficients of added mass and drag	T_{PEP}	Peak energy period (s)
e, f	Local coordinate system fixed with the tube (m)	T_w	Natural period of water column (s)
F_{exc_i}	Matrix of excitation force due to waves (N)	T_Z	Zero up crossing period (s)
F_f	Matrix of quadratic drag force (N)	U	Amplitude of water column displacement (m)
F_m	Matrix of mooring forces (N)	V	Volume of water displaced (m^3)
F_{PTO}	Matrix of damping force due to power take off system (N)	$w, \dot{w}, \ddot{w}$	Displacement (m), velocity (m/s) and acceleration (m/s^2) of water column in body fixed local coordinate system
g	Gravitational acceleration (m/s^2)	$x, \dot{x}, \ddot{x}$	Displacement (m), velocity (m/s) and acceleration (m/s^2) of U-tube in surge in earth fixed coordinate system
G_{ij}	Buoyancy matrix (N/m)	$y, \dot{y}, \ddot{y}$	Displacement (m), velocity (m/s) and acceleration (m/s^2) of U-tube in heave (m), (m/s), (m/s^2) in earth fixed coordinate system
H	Wave height (m)	$\dot{z}, \ddot{z}$	water particle velocity (m/s) and acceleration (m/s^2) due to waves and current in earth fixed coordinate system
H_S	Significant wave height (m)		
K	Stiffness (N/m)	Symbols	
KE	Kinetic energy (J)	τ	Variable for time (s)
KE_h	Kinetic energy of water in horizontal tube (J)	γ	Peakedness parameter
KE_{lv}	Kinetic energy of water in left vertical tube (J)	ρ	Density of water in tube (1000 kg/m^3)
KE_{rv}	Kinetic energy of water in right vertical tube (J)	ρ_w	Density of sea water (1025 kg/m^3)
KE_{str}	Kinetic energy of U-tube (J)	ϕ	Velocity potential (m^2/s)
K_{surge}	Stiffness of U-tube in surge (N/m)	ω	Angular frequency (rad/s)
K_{heave}	Stiffness of U-tube in heave (N/m)	$\xi_j, \dot{\xi}_j, \ddot{\xi}_j$	Displacement (m)/(rad), velocity (m/s)/(rad/s) and acceleration (m/s^2)/(rad/s ²) at jth DOF
K_{pitch}	Stiffness of U-tube in pitch (N/m)	θ_1, θ_2	Inclination of R_1 & R_2 with respect to the local axis f (rad)
L	Total length of water column (m)	Acronyms	
L_A	Lagrangian (J)	BBDB	Backwater Bent Duct Buoy
L_C	Distance of centre of resultant gravity from centreline of bottom tube (m)	COG	Centre of Gravity
L_E	Effective length of water column (m)	CW	Capture Width
L_H	Length of horizontal tube (m)	CWR	Capture Width Ratio
L_T	Distance of centre of gravity of U-tube from centreline of bottom tube (m)	DOF	Degree of Freedom
L_{TV}	Vertical length of tube (m)	MOI	Moment of Inertia
L_V	Vertical length of water column (m)	MSL	Mean Sea Level
m_{ij}	Mass matrix (kg)	MWL	Mean Water Level
M	Mass of U-tube and buoys (kg)	OWC	Oscillating Water Column
M_W	Mass of water column (kg)	PTO	Power Take Off
MI_s	Moment of inertia of U-tube and buoys	TLC	Tuned Liquid Column
MOI_p	Moment of inertia in pitch (kg m^2)	TLCD	Tuned Liquid Column Damper
N_Z	Number of zero upcrossings	TRL	Technology Readiness Level
$p, \dot{p}, \ddot{p}$	In pitch, displacement (rad), velocity (rad/s), and acceleration (rad/s^2) of U-tube	UK	United Kingdom
P_e	Power extracted (Watt)	USD	United States Dollar
PE	Potential energy (J)	WEC	Wave Energy Converter
PE_h	Potential energy of water in horizontal tube (J)		
PE_{lv}	Potential energy of water in left vertical tube (J)		
PE_{rv}	Potential energy of water in right vertical tube (J)		
PE_{str}	Potential energy of U-tube (J)		
q	Vector of degree of freedoms		
Q_{nc}	Vector of non-conservative forces (N)		

type of system can be shore-fixed or floating type. Several review papers (such as [10,11]) are available in the literature summarizing the experience gained on the OWCs since the first large scale deployment of “Kaimei” took place in Japan [12]. Among other applications, the OWC systems used in Toftestallen (Norway [13]), in Vizhinjam (India [4]), in Pico (Portugal [14]), in Islay (UK [15]), in Guangdong (China [16]),

etc., can be mentioned. The major drawback of this system is the requirement of a massive structure at the marine environment that involves large initial investment. Considering this, a few OWCs have been constructed by integrating these into breakwaters in order to share the construction cost, such as Mutriku in Spain [17], U-OWC in Italy [18], etc. However, this approach is not suitable for all types of breakwaters.

Most of these projects are regarded as demonstration projects, and the wave energy concentration effect available at these sites may not be easily replicable at other locations [10]. Some of these shore-based structures could not survive the harsh ocean environment (e.g., OSPREY [10]), indicating the operational difficulties associated with such systems. Regarding the floating OWCs, the concept of Backwater Bent Duct Buoy (BBDB) had been evolved during 1980s [19], where the water column is L-shaped. Since then several studies have been carried out on floating OWCs, including a scaled model test in Ireland [20]. Among various other floating OWC concepts, the Mighty Whale of Japan with multiple air chambers [21], OWC Spar-buoy utilizing a long vertical tube opened at both ends [22], floating WECs incorporating interior OWC [23], etc., can be mentioned. In fact, the OWC Spar-buoy has been regarded as having the lowest risk and highest economy for development in the Atlantic regions by the British Department of Trade and Industry in 2005 [24]. Multiple OWCs under a single structure have also been considered in several studies, such as: Seabreath [25], Leancon [26], device at the University College Cork, Ireland [27], etc. Several other studies have been conducted to enhance the overall efficiency of the OWCs, such as: by using slanted walls instead of vertical one [28], adjusting chamber length according to incident wave period to achieve resonance [29], combining OWC and oscillating buoy as a hybrid system [30], etc. All these techniques are at their various levels of development, and unfortunately the pressure differential based concepts have not been evolved further for large scale commercial application.

In a Floating Structure type device, wave induced relative motion between multiple floaters deployed on the water surface or between floaters with respect to fixed locations is converted into useful mechanical energy. This is then used either to directly generate electricity or to convert into hydraulic head for running a turbine. For example, several long floating cylinders are connected together in the Pelamis device [31], (Fig. 1b), which is moored parallel to the wave direction, and the relative pitch motion between two adjacent cylinders is used to produce power from it. In the point absorber system, the heave motion of a floating buoy relative to a bottom fixed base (or a floating base which is relatively still) is used to extract energy from the waves. The buoy attached to a bottom fixed base is simple to design, but faces problems due to the changing water levels during the tides. The early

models installed utilizing the fixed base concept are the G-1T (Tokyo [32]) supported in a breakwater and the Norwegian buoy with spherical floater [33]. The Bipartite point absorber [34] and the IPS buoy incorporating piston-tube power take off system [35] are the examples of floating point absorbers. Another scaled down model of a floating device, the Wavebob, has been tested in Ireland in the early 2000 [36]. So far, however, the floating structure concept has not been considered for large scale deployment.

In an Overtopping Device (Fig. 1c), the water close to the wave crest is captured as it is spilt over into a reservoir placed at a level higher than the surrounding mean sea level. This water is then used to run a low-head hydraulic turbine [37]. Since the waves are not directly applying forces on the PTO system, the performance of this WEC technology does not depend on establishing a system in resonance with the waves, and therefore provides design flexibility. The whole overtopping set-up can be either bottom fixed or floating type. One of the early bottom fixed overtopping projects is Tapchan [38] in Norway where a collector (to focus and concentrate the waves) and a convertor (to convert energy of waves into potential energy in the reservoir) have been used. In Seawave Slot-Cone Generator [39], a ramp wall has been utilized to create wave run-ups and spill water into a reservoir. This system is integrated into a caisson breakwater. The economic advantage of incorporating such WECs in the coastal protection structures such as rubble mound breakwaters or vertical structures has been reiterated in Ref. [40]. Application of a two stage ramp has been presented in Ref. [41] to increase the overtopping flow rate for such WECs. Apart from the associated cost, a major problem with these structures is the operational efficiency and structural survivability [40]. Among the floating overtopping devices, Wave Dragon in Denmark [42] can be mentioned. A study on a prototype of this overtopping model has been presented in Ref. [43]. The design of such overtopping devices is a challenging task in order to optimize the power capture, and so far none of these have been fully commercialized.

In an Impact Device type WEC, energy is extracted primarily from the surge motion of the water particles. As shown in Fig. 1d, these are usually placed perpendicular to the wave direction, and a force collecting component (e.g., a flap) is moved in the surge direction by the waves. One of the earliest devices under this category is the Mace [44],

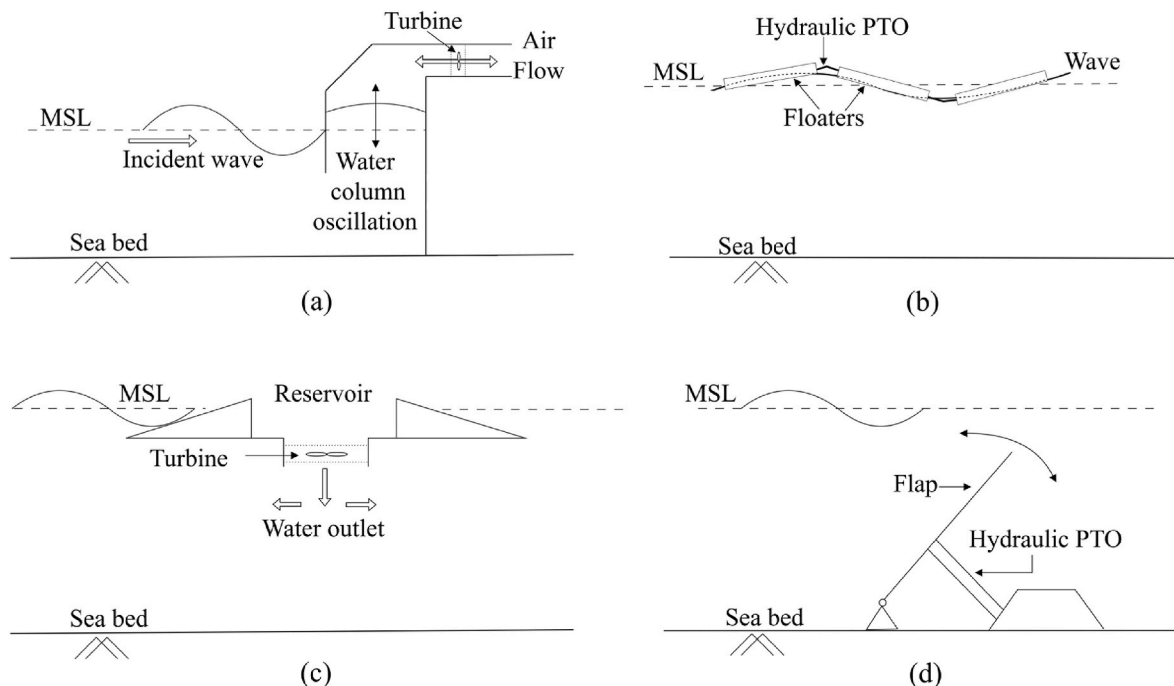


Fig. 1. Schematic diagrams of WEC principles (a) OWC, (b) Pelamis, (c) overtopping, (d) impact.

in which an inverted pendulum cabled with a hydraulic PTO system and hinged at the seabed has been used. Another example of such a device is the Oyster [45] in Scotland whose second generation prototype has been installed at rather shallow waters (13 m). Among other major attempts under this category, the Wave Roller [46], the Biowave [47], and the Langlee [48] can be mentioned. Several other impact type devices with variable geometry and control surfaces are also currently under development [49]. Unfortunately, none of these approaches has been evolved so far into a commercially viable solution for regular power production.

The estimated useable wave power available along the global coastlines is around 2 TW [50]. Since 2010 to 2023, a total of around 25 MW of wave power capacity has been installed for testing and demonstration [51]. In total, only around 400 MW of wave power are currently under development through various projects [52]. However, despite the showcasing of technical feasibility of various concepts through numerous prototypes, the active wave power capacity in 2020 has been as small as 2.31 MW [52]. Growth and commercialization of these systems are still hindered by their massive cost compared to other renewable alternatives [53]. The major reasons of such poor performance of the wave energy sector, despite the boom in renewable energy in today's world, can be summarized as [9].

- Mechanical motions obtained from waves are typically low frequency and high force oscillations, which are difficult to convert into electricity. This often requires robust and reliable structures and PTO systems, i.e., involvement of high initial investment.
- Operational and maintenance expenses are high due to the marine work environment.
- Wave energy resource varies with time and location in terms of frequency, wave height, and direction. This requires detailed investigation in diverse WEC technologies to select the best one for a certain site, which eventually increases the overall cost for research and development, and commercialization.
- The structural failure probability is relatively high in an extreme weather conditions, which makes the investment in WEC technologies riskier.

In order to address some of the above issues, the application of a floating U-tube Tuned Liquid Column (TLC) has been investigated in this paper. TLCs have been widely used as dampers (TLCD) to suppress the vibration of structures (e.g., tall buildings [54], bridges [55], etc.) excited by the wind/seismic loads or the roll motion of ships [56] in waves. In such applications, the vibration of the main structure is suppressed as energy is collected from it and dissipated by the sloshing water column when it passes through an orifice placed inside the tube.

Few researchers have considered this sloshing motion of water in a floating U-tube to extract power from the waves (for e.g., see Refs. [23, 57]). Even though the system involves a water column that oscillates, it is different from the conventional OWC type of devices as the water in the U-tube is not connected to the open ocean, i.e., the system is interior and hence could be robust. Any of the rigid body DOFs of the floater, that can excite the water column, can be used to absorb energy from the waves, and hence it may be expected that a wider range of wave frequencies can be utilized by this system.

An asymmetric floater with an internal U-tube has been considered in the UGEN concept as presented in Ref. [58]. The validation of the numerical model and the dynamic characteristic of the water column have been established therein through experimental and numerical studies carried out on a scaled down model. The size of the prototype floater considered in Ref. [58] is 20m × 15m, with 5m draft, and the total mass of water is 490 ton. The oscillating water surface has been considered to force the air above the water surface to pass through an air-turbine for generating power. In the numerical study in Ref. [58], the motion of the liquid column has been modelled using one dimensional Euler equation (as available in Ref. [59]) assuming small rotational motion of the water column. The PTO has been modelled using a

quadratic damping term. One major conclusion drawn in Ref. [58] is, that the motion of the water column can be amplified at two separate frequencies: one corresponds to the water column motion in the U-tube and the other to the floater's roll motion. This observation is typical to a 2 DOF spring-mass system. In another work involving the same floater, as presented in Ref. [23], the dynamics of the PTO system has been modelled with a differential equation involving the air pressure fluctuation. The peak efficiency for the system considered therein has been reported to be around 0.47, which is lying between wave heights 2–3 m and around the roll natural period of the floater with a narrow bandwidth. In Ref. [57], a cross flow water turbine utilizing the bi-directional flow of a U-tube water column has been studied experimentally and numerically, where 35–45 % of maximum hydraulic efficiency has been reported. The geometry of the horizontal segment of the U-tube has been designed such that even the bi-directional water flow can maintain unidirectional rotation of the turbine.

It can be noticed from the literature survey on the U-tube floaters that all the previous works are oriented primarily to establish the characteristics of a U-shaped water column and the PTO system. The authors have not come across any investigation considering an actual deployment of such a system at a site. In the current study, the deployment of a circular cross-section U-tube WEC unit, for producing around 50 kW power, afloat by a pair of buoys (Fig. 2b) and moored to the seabed has been considered. It is expected that the application of multiple such units can reduce the initial cost involved with the development of wave energy. The sensitivity of the extracted energy with respect to the major parameters defining the floating U-tube structure has been investigated. The performance of the proposed mooring system has been checked for an extreme sea-state considering a certain site. The results presented here could form the basis for an actual design of such a system. In the following sections, first the proposed system has been described including the major parameters that can impact its performance. Thereafter the mathematical model used in the analysis has been presented in brief. Next, the validation of the model has been carried out by comparing with the results of simpler TLC models which are well established in the literature. Finally, the results obtained from the sensitivity analysis and the survival condition analysis have been summarized and conclusions are drawn.

2. Proposed U-tube water column for extracting wave energy

A schematic sketch of the floating U-tube is shown in Fig. 2a. It is assumed to be made of steel, while the buoys will be made of standard polypropylene/polyethylene based materials commonly used by the offshore industry in buoyancy modules. The U-tube cross-section is assumed to be circular and uniform throughout its length, while two of hydraulic turbines connected to two of generators will be used as the PTO to extract power from the oscillating water column as depicted in Fig. 2a. The set of preliminary dimensions assumed for the system is presented in Table 1. The wave period of the prevailing sea-state (operating condition) has been taken as 7 s. At many coastal locations around the world, the maximum annual wave energy has been observed to exist around this wave period (e.g., see Ref. [60]). Even though six rigid body degrees of freedom are associated with a floating body, currently it is assumed that the U-tube will be aligned along the wave directions, and hence only surge, heave and pitch motions of the structure will be important for exciting the water column. The natural period of an oscillating water column (T_w) in a U-tube can be expressed as in Eq. (1) [61]. It is evident that resonating oscillation of the water column is required to extract maximum amount of energy from the waves.

$$T_w = 2\pi \sqrt{\frac{L_E}{2g}} \quad (1)$$

where, L_E = effective length of water column = $2L_V + \frac{A_V}{A_H}L_H$, (major

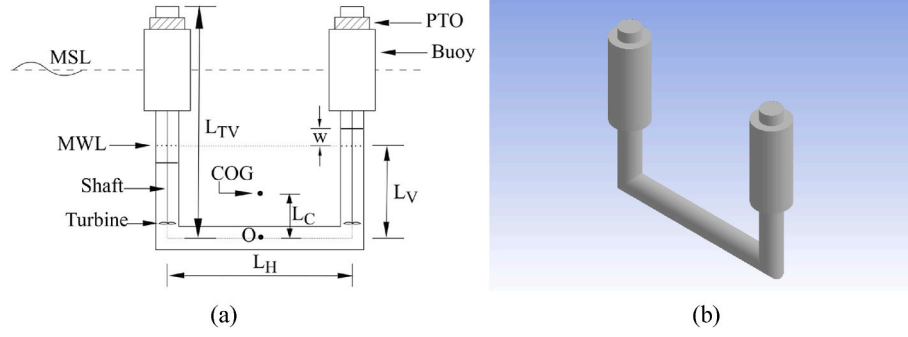


Fig. 2. (a) Schematic diagram of U-tube WEC, (b) Model in a hydrodynamic tool.

Table 1

Major dimensions of U-tube WEC.

Parameters	Value
Length of Vertical, horizontal tubes (L_{TV}, L_H) ^a	15 m, 16 m
Outer diameter and thickness of all tubes	2 m and 10 mm
Vertical water column (L_V) ^a	4 m
Outer and inner diameter of buoys	3.5 m and 2 m (total available volume 154 m ³)
Material of tubes	Steel (density 7850 kg/m ³)
Density of material used in the buoys	300 kg/m ³
Total mass of horizontal and vertical tubes	54.1 ton
Mass of water column	73.9 ton (density 1000 kg/m ³)
Location of centre of gravity (COG) of overall system (L_C) ^a	3.73 m
Available free height above mean water surface inside tube	11 m

^a Marked in Fig. 2a.

terms are shown in Fig. 2a), A_V = area of vertical tube, A_H = area of horizontal tube, L_V = length of water in vertical tube, L_H = length of horizontal tube.

For tuning the water column's natural period to a wave period of 7 s (with an initial assumption of $\frac{A_V}{A_H} = 1$), the total water column length can be estimated as 24 m using Eq. (1). It may be mentioned here that the tuning of a mass damper system using the mass ratio, as may be found in Ref. [62], has not been attempted here. The U-tube has been assumed to be rigid in the current study and no structural analysis of the tube has been carried out. For the sensitivity study, four (4) major parameters considered are:

1. Damping coefficient (C)
2. Ratio of the vertical to the horizontal water column length $\left(R_L = \frac{L_V}{L_H}\right)$.
3. Ratio of the vertical to the horizontal tube cross sectional area $\left(R_A = \frac{A_V}{A_H}\right)$.
4. Ratio of the mass of water to the total mass of the buoys and the steel in the tubes $\left(R_M = \frac{M_w}{M}\right)$. During these parametric variations, the effective length of water column has been kept constant.

As pointed out in Ref. [58], in the frequency response of the water column of such a system, two peaks are visible - one corresponds to the natural period of the water column and the other to the pitch natural period of the floating structure. This is a common feature of a two DOF spring-mass system (such as, see Ref. [62]). It can be assumed that the presence of two peaks in the response spectrum could be beneficial to extract power from a wider wave period range. This aspect, however,

has not been investigated in the current study.

The natural periods of a floating structure in pitch (T_p) and heave (T_h) can be expressed as in Eq. (2) (any hydrodynamic effect can be assumed to be included in the mass and stiffness terms) [63].

$$\left. \begin{aligned} T_p &= 2\pi \sqrt{\frac{Mol_p}{\rho_w g A_b (L_H)^2 / 2}} \\ T_h &= 2\pi \sqrt{\frac{M}{\rho_w g A_b}} \end{aligned} \right\} \quad (2)$$

Where, M, Mol_p are mass and mass moment of inertia in heave and pitch, ρ_w = density of sea water, g = acceleration due to gravity, A_b = total water plane area, L_H = length of the horizontal tube.

It is obvious that the tuning of the water column is much easier as it depends on the length of the water column, and the ratio of the vertical to the horizontal tube cross-sections. On the other hand, the pitch natural period of the structure is dependent on the mass and geometry of the U-tube structure, the buoys, and the mooring system, and hence the freedom to tune the pitch natural period is relatively restricted. Hence, with this practical consideration, no attempt has been made currently to control the pitch natural period of the system during the sensitivity studies.

For station keeping of the structure at the assumed water depth of 50 m, a spread mooring system (for e.g., see Ref. [64]) consisting of two lines connected at the level of the system's COG (Fig. 3b) has been considered along with one clump weight attached to each of the lines. Details on the mooring lines have been presented in Table 2. This mooring system has been considered after carrying out a few random trials with different configurations; however, optimization of the mooring system has been excluded from the scope of the current study. It may be mentioned that the total weight of the mooring lines must be within the buoyancy available from the buoys such that the designed draft can be maintained. It has been noticed during the simulations that the structure oscillates about a drifted mean position, as shown in Fig. 3b, due to the inclusion of the wave drift forces in the calculations. No mooring lines have been used currently along the sway direction.

The water depth and the environmental conditions has been referred from Ref. [60] related to a location of the Indian coastal region. These are shown below in Table 3. The extreme sea-state of the same site has been referred from Ref. [65]. Any effect due to the ocean current has been ignored in the present study.

3. Mathematical model of the U-tube WEC

As already mentioned, it is assumed that the water column motion is coupled with surge, pitch, and heave of the U-tube. The associated problem has been solved by coupling two models - one for the floating body and the other for the water column - through a dynamic link library (dll) file written in FORTRAN. Forces acting on the floating body due to the movement of the water column have been calculated inside the dll

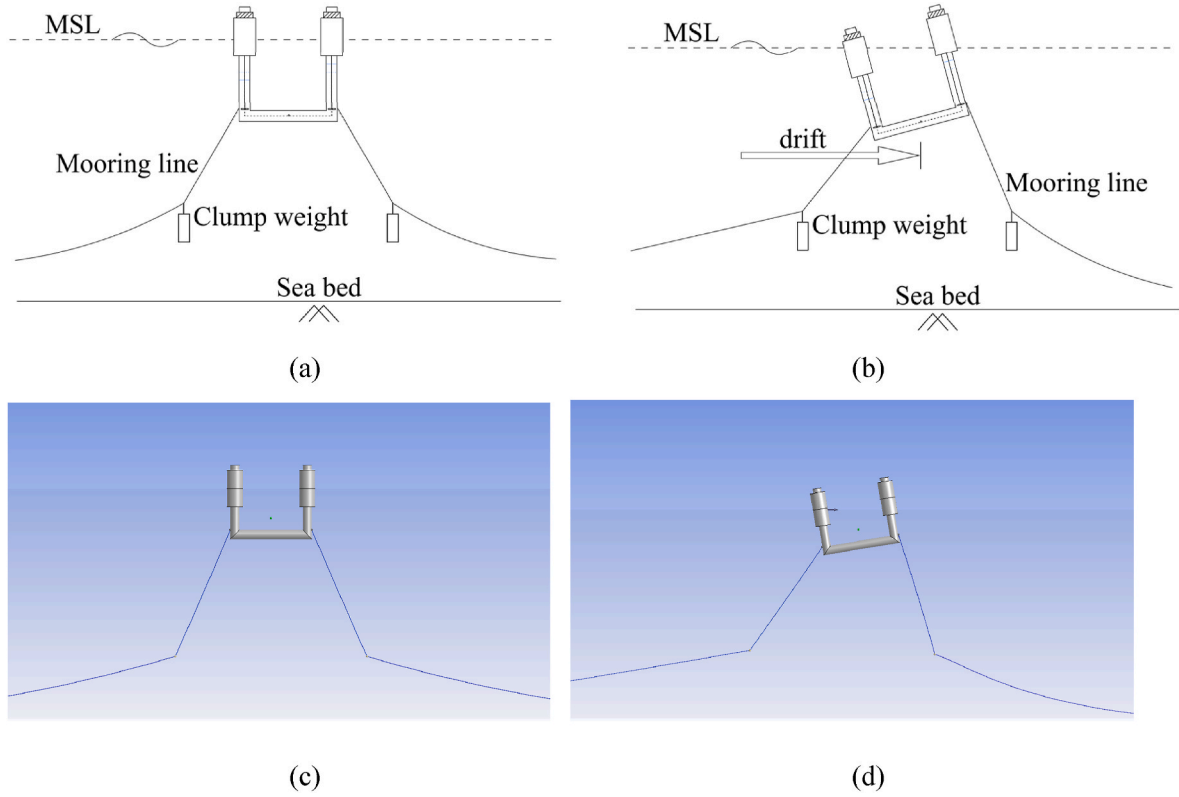


Fig. 3. Mooring system of U-tube WEC (a) and (c) no wave condition, (b) and (d) oscillating at a mean drifted position.

Table 2
Details of the mooring system.

Parameters	Value
Diameter of mooring cable	25 mm
Material used in mooring cable	Steel wire rope
Mass per unit length of cable	3.13 kg/m
Axial stiffness of cable	36 MN
Minimum Breaking Force of cable	576 KN
Clump weight on each cable	2000 kg

Table 3
Site condition and environmental condition.

Parameters	Value
Water depth	50 m
Operating condition wave height and period	1 m, 7 s
Extreme wave height, wave period (regular waves)	7.6 m, 8.3 s (return period of 20 years) ^a
Extreme H_s , T_{PEP} , γ (irregular waves)	4.4 m, 9.03 s, 2.25 ^a

^a For the Indian coastal region, ref. [65].

file and applied as external forces acting at the COG of the floating body.

The generalized form of the equation of motion of the system can be modelled using Cummins approach [63] which is shown in Eq. (3).

$$\sum_{j=1}^n (m_{ij} + A_{\infty ij}) \ddot{\xi}_j(t) + \sum_{j=1}^n \int_0^{\infty} B_{ij}(t-\tau) \dot{\xi}_j(\tau) d\tau = F_{exc_i}(t) - \sum_{j=1}^n G_{ij} \xi_j(t) - F_m(\xi_j, \dot{\xi}_j, \ddot{\xi}_j) - F_f(\xi_j) - F_{PTO}(\xi_j, \dot{\xi}_j) \quad (3)$$

Where, ξ_j , $\dot{\xi}_j$, $\ddot{\xi}_j$ are the displacement, velocity and acceleration at jth degree of freedom respectively, m_{ij} is the mass matrix of the system, $A_{\infty ij}$ is the matrix of so called added mass infinity, B_{ij} is the added matrix due

to radiated waves (retardation function), F_{exc_i} is excitation force due to waves, G_{ij} is the buoyancy matrix, F_m is mooring forces, F_{PTO} is damping force due to power take-off system, F_f is the matrix of damping forces due to flow separation (quadratic drag force).

The hydrodynamic forces acting on the mooring lines can be estimated using Morison equation, as shown in Eq. (4).

$$F_m(t) = (1 + C_A) \rho_w V \ddot{z} - C_A \rho_w V \ddot{x} + \frac{1}{2} C_D \rho_w A (\dot{z} - \dot{x}) |\dot{z} - \dot{x}| \quad (4)$$

where, ρ_w is seawater density, $\dot{z}$ and $\ddot{z}$ are the water particle velocity and acceleration (including current), $\dot{x}$ and $\ddot{x}$ are the structure's velocity and acceleration, C_A , C_D are added mass and drag coefficients. These can be obtained for a circular section from literature (such as DNV-RP-H103 [66]).

In the current study, Ansys-AQWA has been used first to obtain the added mass and linear (potential) damping, the quadratic transfer function, and thereafter to carry out the time domain simulation of the system (i.e., integration of Eq. (3)) [67]. For the hydrodynamic analysis, in total, 15140 quadrilateral elements have been used (maximum size of 0.35 m) in the wetted surface. The viscous (quadratic) damping coefficient matrix for the structure have been approximated by assuming flow perpendicular to the member axis and using the standard drag coefficients for a straight cylinder as available in Ref. [66]. The panel size in the hydrodynamic analysis and the time step used in the subsequent time domain simulations (which is 0.01 s) have been chosen after conducting a convergence study.

The force applied by the movement of the water column at each time step has been added to the above system as an external force (i.e., F_{PTO} of Eq. (3)) using a dll file in which the mathematical model of the water column movement has been implemented. A flow chart explaining the numerical approach is depicted in Fig. 4.

Since the diameter of the tubes (of the PTO) are much smaller compared to its length, the equations of motion of the water column

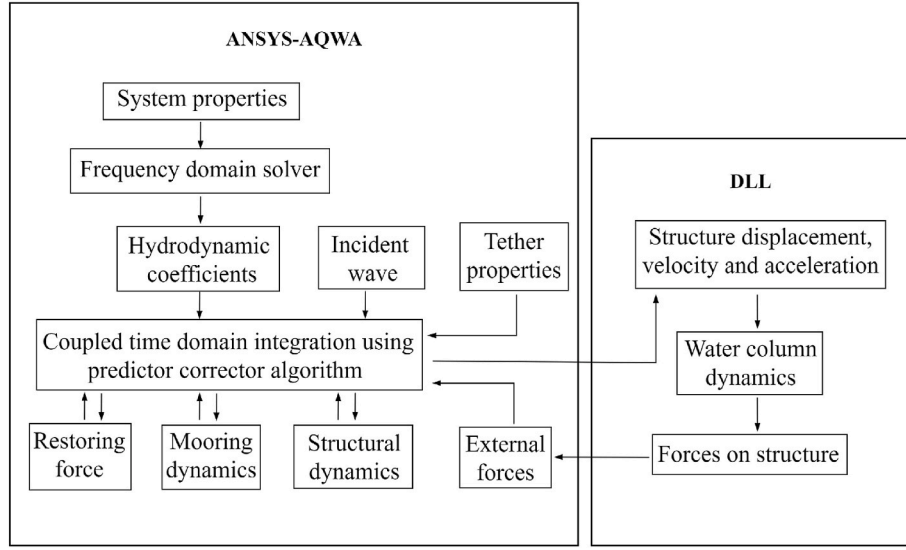


Fig. 4. Flow chart of coupled analysis between the hydrodynamic model in Ansys-AQWA and U-tube model in dll.

have been obtained using Lagrangian mechanics as expressed in Eq. (5) [68,69].

$$\frac{d}{dt} \left(\frac{dL_A}{dq} \right) - \frac{\partial L_A}{\partial q} = Q_{nc} \quad (5)$$

Where, $L_A = KE - PE$.

In the above, L_A is the Lagrangian, KE, PE are the total kinetic and potential energy of the system respectively (which are shown in Eq. (6)), q is the vector of the degrees of freedom, Q_{nc} is the vector of the total non-conservative forces.

In Eq. (6), the velocities of water in the tube segments used to calculate the kinetic energy have been measured with respect to the centre of the horizontal tube “O” in the local coordinate system as shown in Fig. 5a, while the total potential energy has been calculated with respect to the earth fixed coordinate system. Also the surge, pitch and heave motions of the whole floating body are defined in the earth fixed coordinate system.

$$\begin{aligned} KE &= KE_{lv} + KE_{rv} + KE_h + KE_{str} \\ PE &= PE_{lv} + PE_{rv} + PE_h + PE_{str} \end{aligned} \quad (6)$$

Where,

$$\begin{aligned} KE_{lv} &= \frac{1}{2} \rho A \int_0^{L_v-w} \{ \dot{x} \cos p + \dot{y} \sin p - (f - L_c) \dot{p} \}^2 df \\ &+ \frac{1}{2} \rho A \int_0^{L_v-w} \left(-\dot{x} \sin p + \dot{y} \cos p - \frac{L_H}{2} \dot{p} - \dot{w} \right)^2 df \end{aligned} \quad (7)$$

$$\begin{aligned} KE_{rv} &= \frac{1}{2} \rho A \int_0^{L_v+w} \{ \dot{x} \cos p + \dot{y} \sin p - (f - L_c) \dot{p} \}^2 df \\ &+ \frac{1}{2} \rho A \int_0^{L_v+w} \left(-\dot{x} \sin p + \dot{y} \cos p + \frac{L_H}{2} \dot{p} + \dot{w} \right)^2 df \end{aligned} \quad (8)$$

$$\begin{aligned} KE_h &= \frac{1}{2} \rho A \int_{-\frac{L_H}{2}}^{\frac{L_H}{2}} (\dot{x} \cos p + \dot{y} \sin p + L_c \dot{p} + \dot{w})^2 de \\ &+ \frac{1}{2} \rho A \int_{-\frac{L_H}{2}}^{\frac{L_H}{2}} (-\dot{x} \sin p + \dot{y} \cos p - e \dot{p})^2 de \end{aligned} \quad (9)$$

$$\begin{aligned} KE_{str} &= \frac{1}{2} M \{ \dot{x} - (L_T - L_c) \dot{p} \cos p \}^2 + \frac{1}{2} M \{ \dot{y} - (L_T - L_c) \dot{p} \sin p \}^2 \\ &+ \frac{1}{2} M I_s \dot{p}^2 \end{aligned} \quad (10)$$

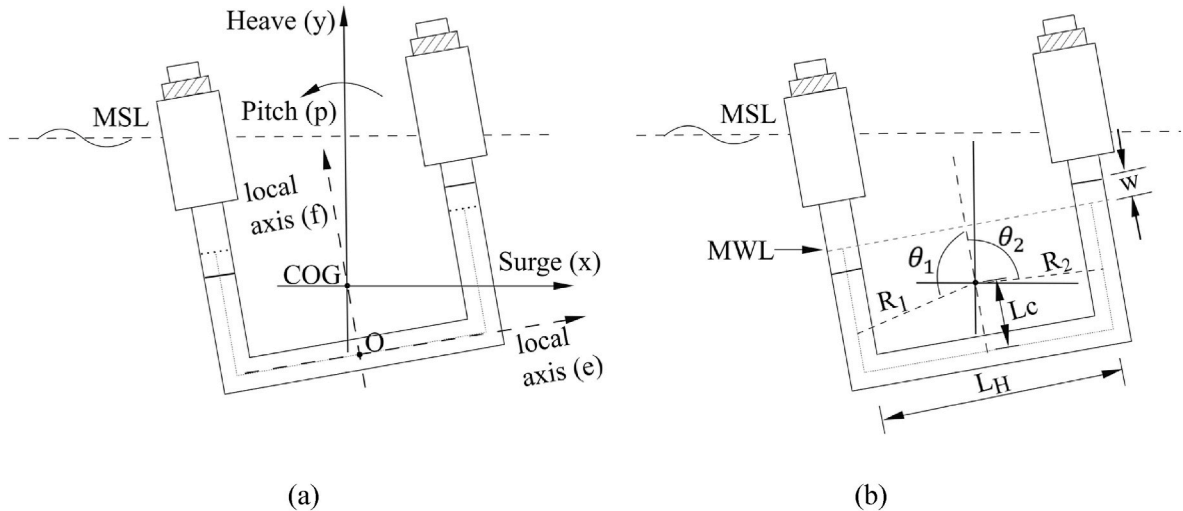


Fig. 5. (a) Schematic diagram showing DOFs, (b) Schematic diagram showing details of coordinate system.

$$PE_{lv} = \rho g A (L_v - w) (L_C + R_1 \cos(\theta_1 + p) + y) \quad (11)$$

$$PE_{rv} = \rho g A (L_v + w) (L_C + R_2 \cos(\theta_2 - p) + y) \quad (12)$$

$$PE_h = \rho g A L_H (L_C (1 - \cos p) + y) \quad (13)$$

$$PE_{str} = Mg \{L_C + (L_T - L_C) \cos p + y\} + \frac{1}{2} K_{surge} x^2 + \frac{1}{2} K_{heave} y^2 + \frac{1}{2} K_{pitch} p^2 \quad (14)$$

In the above,

KE_{lv}, KE_{rv}, KE_h = kinetic energy (KE) (J) of water in left vertical, right vertical and horizontal tubes,

PE_{lv}, PE_{rv}, PE_h = Potential energy (PE) (J) of water in left vertical, right vertical and horizontal tubes,

KE_{str}, PE_{str} = KE (J) and PE (J) of U-tube,

ρ, A, g = water density (kg/m^3), cross-sectional area of tube (m^2) and gravitational acceleration (m/s^2) respectively,

$x, \dot{x}, \ddot{x}$ = displacement (m), velocity (m/s), and acceleration (m/s^2) of the U-tube in surge,

$w, \dot{w}, \ddot{w}$ = displacement (m), velocity (m/s), and acceleration of the water column,

$y, \dot{y}, \ddot{y}$ = displacement (m), velocity (m/s), and acceleration (m/s^2) of the U-tube in heave,

$p, \dot{p}, \ddot{p}$ = displacement (rad), velocity (rad/s), and acceleration (rad/s^2) of the U-tube in pitch,

e, f = Local coordinate system fixed with the tube (m),

L_H, L_C, L_V = length of horizontal tube (m), distance of centre of resultant gravity from centreline of bottom tube (m), length of vertical water column respectively (m),

L_T = distance of centre of gravity of U-tube from centreline of bottom tube (m),

M, I_s = mass and moment of inertia of U-tube structure,

R_1, R_2 = distance of centre of gravity of water column in left vertical tube and right vertical tube respectively from the COG of complete system (m),

θ_1, θ_2 = inclination of R_1 & R_2 from local f axis (rad)

From the equations of motion obtained using Eq. (5), terms representing the forces acting on the structure along its surge, pitch and heave degrees of freedom due to the water column's kinematics are collected. These are evaluated at each time step and applied at the COG of the structure as external forces through the dll file. These terms are presented in Appendix A.

4. Numerical validation of the model using simpler cases and calculation of its natural periods

The problem of the motion of water column in a U-tube has been widely investigated by the TLCD research community. This includes numerical as well as experimental studies on the U-tube undergoing surge [70] and pitch [71] motions individually, as well as in combination [69]. As already mentioned, these models have been developed for suppressing vibration of structures in either one or two DOFs, but they lack coupled analysis in surge, heave, and pitch DOFs necessary for representing a floating WEC. Nonetheless, these mathematical models of the primary structure either in surge or in pitch are well known in the literature and easier to derive, and can be integrated in time domain using tools like Matlab [72]. In the current study, these simpler and well-known solutions have been used to validate the current mathematical model of the full system (i.e., floating U-tube in Ansys-AQWA having water column coupled with 3 DOFs of the tube through a dll file) by adopting the procedure explained below. No experimental verification has been attempted in the current study.

At first, a sinusoidal excitation force (see Table 4) has been applied, for example, at the surge DOF of the complete system, and the motion amplitudes of the U-tube in surge (see Table 4) and the liquid column are

Table 4

Comparison of the water column motions obtained from the 4 DOF and the 2 DOF systems.

DOF	Force applied	U-tube motion Amplitude	Water column motion amplitude	
			Full 4 DOF system	Simpler 2 DOF system
Heave	$10^6 \sin(0.5t)$ N	1.02 m	0.00 m	0.00
Pitch	$10^6 \sin\left(\frac{2\pi}{7}t\right)$ Nm	12.43°	3.289 m	3.293 m
Surge	$5 \times 10^4 \sin\left(\frac{2\pi}{7}t\right)$ N	0.206 m	0.742 m	0.745 m

obtained. Next the motion of the U-tube thus obtained has been applied to excite the well-known 2 DOF version of the model (i.e., model containing only surge DOF and water column motion [70]) using Matlab in time domain. Finally, the motions of the water column thus obtained from these two above models have been compared, as shown in Fig. 6. The same exercise has been carried out for the heave and pitch (using [71]) motion as well, and the results are shown in Table 4 and Fig. 6.

It can be seen in Fig. 6 that for excitation in the heave DOF, the displacement of the water column is zero. This is obvious since the heave motion does not excite the water column at all and there is zero initial movement of the water column. For excitations along the surge and pitch DOFs, the estimated motions of the water columns obtained from the complete model and the simpler 2 DOF models in Matlab shows excellent agreement. Hence it has been concluded that the derived mathematical model of the full system is working properly and further studies on the system has been carried out using the same model.

The natural periods of the current system have been estimated using the free vibration analysis (see Fig. 7 for pitch DOF) so that the effect due to the mooring lines can be included. While carrying out the free vibration study for a certain DOF, all others have been kept deactivated.

This free vibration study has been carried out for both: non-drifted as well as drifted conditions and the results are shown in Table 5. It can be noticed that only the surge natural period is affected by the asymmetry of the mooring line due to the drifted position. However, the time period values in surge are not close to the wave excitation range, even for an extreme condition, i.e., no large dynamic amplification in the mooring line tension is expected.

5. Sensitivity study on the power output due to various parameters

The structural parameters considered to study the sensitivity of the power production are mentioned in Section 2 (C, R_L, R_A, R_M), while the initial values of various system dimensions are shown in Table 1. For the operating condition (Table 3), the assumed wave period is 7 s, and the water column length has been decided based on this. Regular (monochromatic) waves approaching from the head sea direction (i.e., along the horizontal tube) have been considered during all sensitivity as well as extreme condition studies. For most of the sites with significant wave energy available throughout the year, the extreme sea state should be studied with irregular waves as well, as results obtained from regular waves could be overly conservative.

The PTO has been modelled as a linear damping system. The simulation duration has been chosen such that at least 10 cycles of oscillation have been included and the response amplitude has been read after excluding the transient region. It has also been mentioned earlier that during the operating/extreme condition, the structure will be drifted from the zero mean position and will oscillate about a neutral position based on the mooring line configuration. For this reason, in fact, the mean pitch of the structure will be non-zero at higher wave heights, i.e., the structure will exhibit small amount of trim, which can affect the performance of the system to some extent, particularly if the mass of

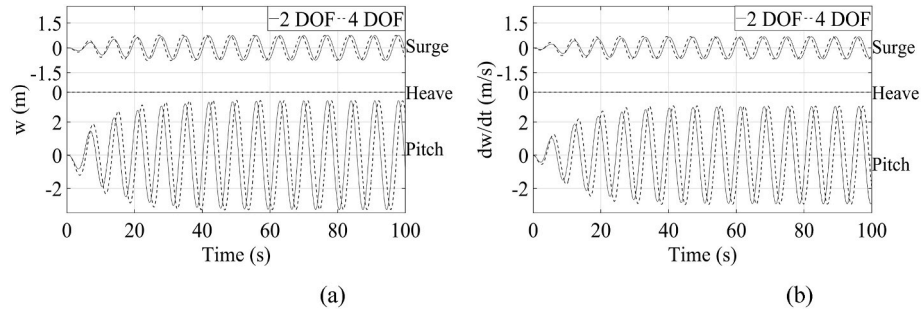


Fig. 6. Comparison of water column motions (excited by surge, heave and pitch of the U-tube) between the full system and 2 DOF systems, (a) displacement w , (b) velocity $\dot{w}$.

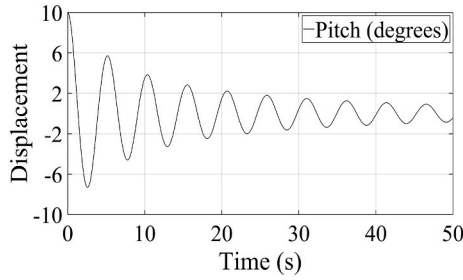


Fig. 7. Free vibration analysis in pitch.

Table 5

Natural periods of the floating U-tube including mooring system.

Parameters	Value
Natural periods of tube in heave, pitch, surge	6.3 s, 6.0 s, 66.7 s
Drifted condition natural periods of tube in heave, pitch, surge	6.4 s, 6.02 s, 38.7 s

water is large.

Modelling the PTO as a linear damper, the power extracted from the system has been estimated from the motion amplitude of the water column using Eq. (15) below [73].

$$\text{Power extracted } (P_e) = \frac{\text{work done by damping force in a cycle}}{\text{time period}} = \frac{2C\pi^2 U^2}{T^2} \quad (15)$$

To describe the hydrodynamic performance of the WEC model under varying geometry and sea states, following indexes are used [74]:

$$\begin{aligned} \text{Capture width (CW)} &= \frac{P_e}{\text{Power available per unit crest length}} \\ &= \frac{P_e}{\rho_w (gH)^2 T / (32\pi)} \end{aligned} \quad (16)$$

$$\begin{aligned} \text{Capture width ratio (CWR)} &= \frac{CW}{\text{characteristic length of WEC}} \times 100 \\ &= \frac{CW}{L_H} \times 100, (\text{in } \%) \end{aligned} \quad (17)$$

Where, ρ_w is density of seawater, g is gravitational acceleration, H is wave height, T is wave time period, L_H is length of horizontal tube, U is the amplitude of water column displacement and C is linear damping constant representing the PTO system.

The CW values represent the width of wave crest whose energy is fully extracted by the system. The CWR can be interpreted as the energy of a single capture width of wave crest absorbed by per unit length of the WEC. It represents only the hydrodynamic efficiency of the system. Even though the equations of motion represent a nonlinear system with

coupled DOFs, a basic requirement for a designer would be an approximately linear (or constant) behaviour of the system, at least within the operational range, in order to design a reliable power source.

Modelling the PTO with linear damping assumption is often adopted to simplify the analysis (such as, see Refs. [75,76]). In this approach, it is important to obtain an idea on the critical damping of the system. This has been obtained by carrying out free vibration study of the system presented in Table 1 with different damping constants, and noting the movement of the water column with time. The results are shown in Fig. 8, which points out that a damping value of 130 kN/m can be regarded as the critical damping of the system. It is evident that too large or too low damping will not be able to produce the desired motion characteristics of the water column to generate appreciable amount of power.

The variation of the extracted power with different damping values is shown in Fig. 9a below. It is noticed that the lines are having larger curvature at higher wave heights. The peak power extraction changes with the damping value for different wave heights. The plots of CW and CWR in Fig. 9b follow a similar pattern for respective wave heights. However, it can be noticed that with increasing wave height, the CW value decreases, pointing out that although the extracted power increases with wave height, an increasing fraction of power goes untapped. The CWR values follow the same trend as CW since L_H remains constant for all the cases.

The operational wave height of the device has been considered to remain in the range of 1–2 m. Hence it is assumed that for the current device, the average of the two damping values corresponding to the peak power extractions for 1m and 2m wave heights, which is = 17.5 kN/m, will be used. All the following sensitivity studies have been carried out by using this damping value.

The variation of the power output with respect to R_L keeping the total length of the water column and all other parameters as constants is shown in Fig. 10a. It can be observed that the structural natural period in pitch almost remains constant due to this variation. The power output reduces gradually (almost linearly) as the ratio increases, i.e., the system will extract more power when the length of the horizontal arm is larger. The reason behind this can be explained considering the excitation mechanism of the water columns by the motions in surge and pitch

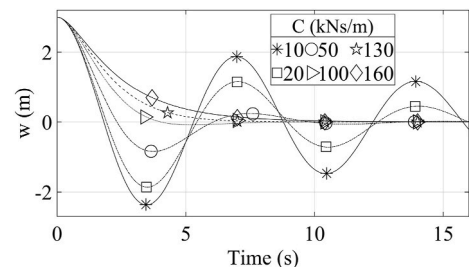


Fig. 8. Free vibration of water column at various damping level.

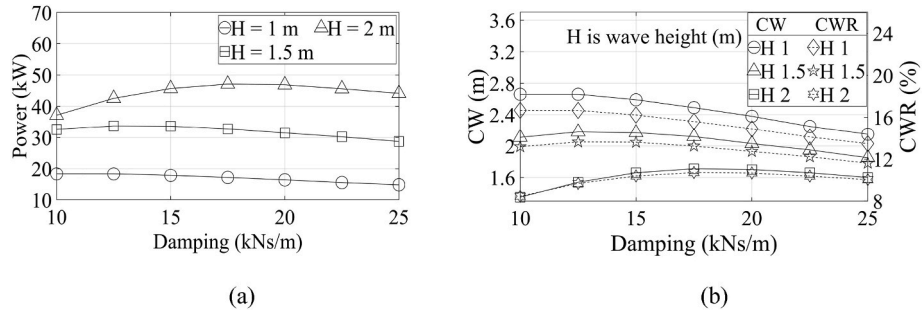


Fig. 9. (a) Power output versus damping applied, (b) CW and CWR versus damping applied.

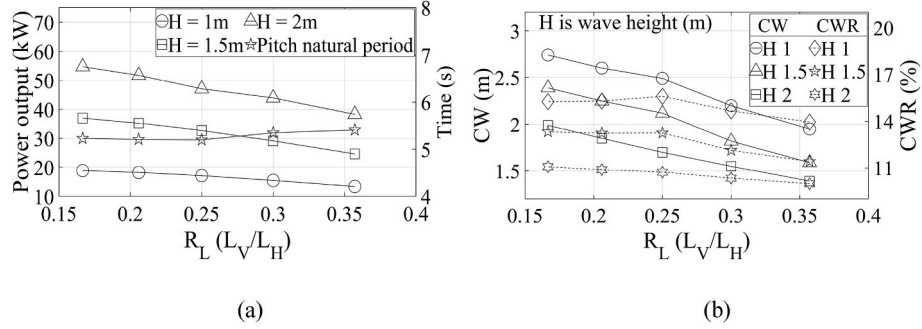


Fig. 10. (a) Power output versus R_L , (b) CW and CWR versus R_L .

DOFs. The surge motion directly excites the water in the horizontal arm, while the pitch motion excites the water in the vertical columns. It is evident that both of these become more effective when the columns are further apart. Hence, the total excitation becomes more intense when

the length of the horizontal arm is larger. Also, it can be noticed that the effect is more prominent at the higher wave heights as the energy in the environment is proportional to the square of the wave amplitude.

Similar to the previous plots, the CW and CWR values decrease with

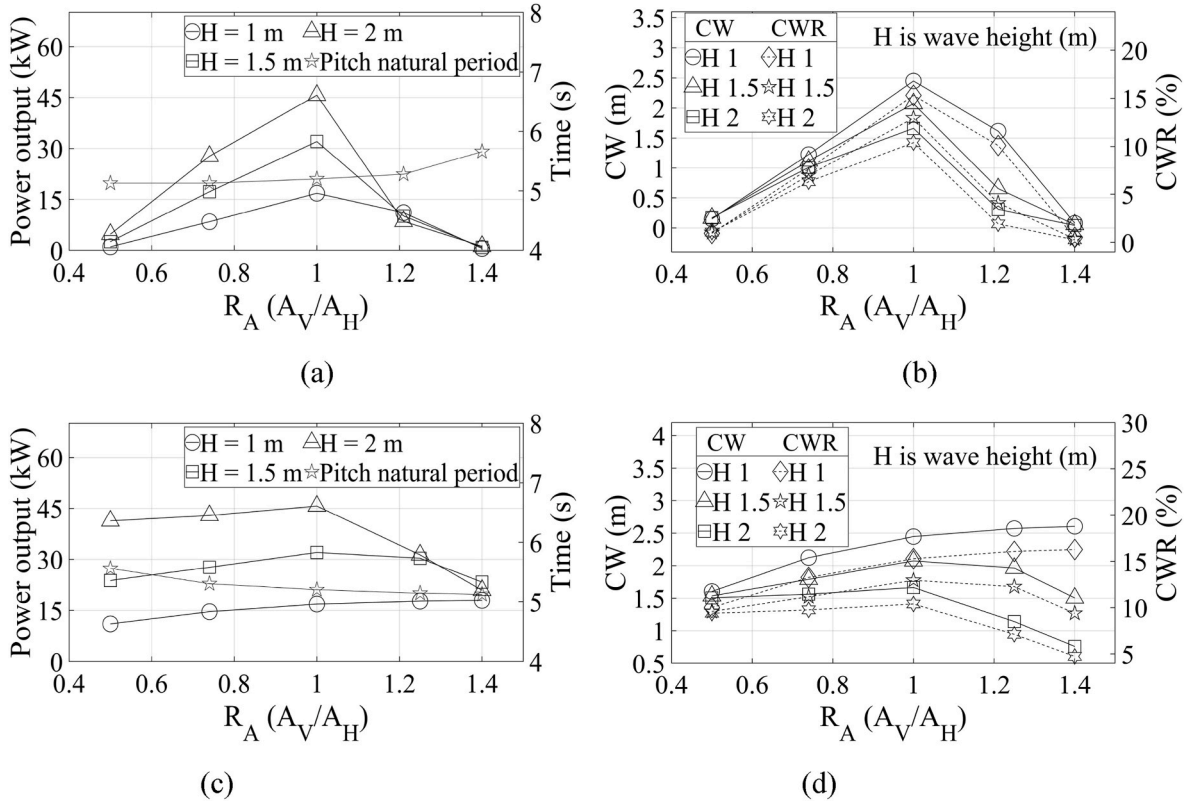


Fig. 11. With variation in vertical tube cross-sectional area: (a) Power output versus R_A , (b) CW and CWR versus R_A ; with variation in horizontal tube cross-sectional area: (c) Power output versus R_A , (d) CW and CWR versus R_A .

increasing wave heights. The CW values decrease almost linearly with increasing R_L , i.e., the system performs better for larger L_H . The CWR value is almost linearly decreasing with higher R_L at higher wave height (2.0 m). However, for lower wave heights, it shows a peak, which coincides with the R_L of the base case structure shown in Table 1. Since CWR only represents the hydrodynamic efficiency, in order to capture more amount of power, a lower R_L value should be preferred.

The variation of the power output with respect to R_A is presented below in Fig. 11. This variation has been achieved by changing the diameters of the tubes keeping the thicknesses constant. It has been implemented by two approaches, first by varying the diameter of the vertical tubes keeping that of the horizontal tube as constant (Figs. a, b), and later by considering the vice-versa (Figs. c, d). The natural periods of the water column, as obtained from Eq. (1), have been kept constant in both approaches by adjusting the length of water column in the vertical arms (L_V).

It may be mentioned here that the practical limitation of L_V should be considered such that the water level neither go above of the U-tube's upper edge, nor below the turbine level. It can be noticed in Fig. 11 that in both the approaches, the maximum power extraction appears at $R_A = 1$, which reduces as the area ratio varies. This is more prominent at higher wave heights. This can be attributed to the fact that the damping has been selected for these simulations following the results of the base case with $R_A = 1$ (Table 1). The natural frequency in pitch DOF of the U-tube remains almost constant throughout in both cases, pointing out that the observed behaviour is primarily due to the varying area ratio.

Similar to previous case, the CW and CWR plots in Fig. 11b show decreasing trend with increasing wave heights. The peaks appear in these plots at $R_A = 1$. On the other hand, the same plots in Fig. 11d show similar trend with smaller curvatures, pointing out the system to be less sensitive to the variation of the horizontal tube's cross-sectional area.

The power output variation with respect to R_M keeping other parameters constant has been shown in Fig. 12. It is evident that the

excitation force of the system primarily depends on the external diameters and lengths of the buoys and the tubes, since these directly interact with the waves. The variation of the ratio has been achieved by two approaches, first by reducing the tube wall thickness keeping the external diameter constant, and thereafter by changing the diameter keeping the wall thickness constant. For the first approach, it can be seen from Fig. 12a that the system natural period in pitch is not varying drastically with R_M . Since the mass and hence the mass moment of inertia in pitch is slowly increasing (compared to the starting case), keeping the excitation force unchanged, the extracted power gradually falls down, which is more prominent at higher wave heights. For the lower wave heights, however, the power production is remaining almost unchanged. The trend of CW and CWR plots in Fig. 12 b follows the extracted power plot closely. For lower wave heights, the values are almost straight line, which changes for higher wave heights and reduces as the ratio increases. Hence it can be concluded that the addition of extra weights due to any accessories of the PTO system etc. may not change the system behaviour abruptly for the lower wave heights.

On the other hand, if the R_M has been changed by increasing the diameter, the nature of the curves changes, and significant change in the energy output can be noticed, as plotted in Fig. 11c and d. The change in the pitch natural period is larger compared to the previous case, but not very significant. Since the diameter is changed, the excitation force and the total mass of water changes significantly. However, as the combined effect, it may be seen that the peak power output has not been changed much, rather falls at higher (or lower) R_M values. At lower R_M values the extracted power is less, indicating that the energy absorbed by the water column is less as its mass is less. The maximum power extraction appears at around $R_M = 1.4$, which is the ratio of the current base case structure (Table 1) for which the maximum energy can be extracted for the chosen damping value. At higher R_M values, the total mass of water is higher, but the power production falls, pointing out insufficient excitation of the water column. The CW and CWR values show similar pattern, showing

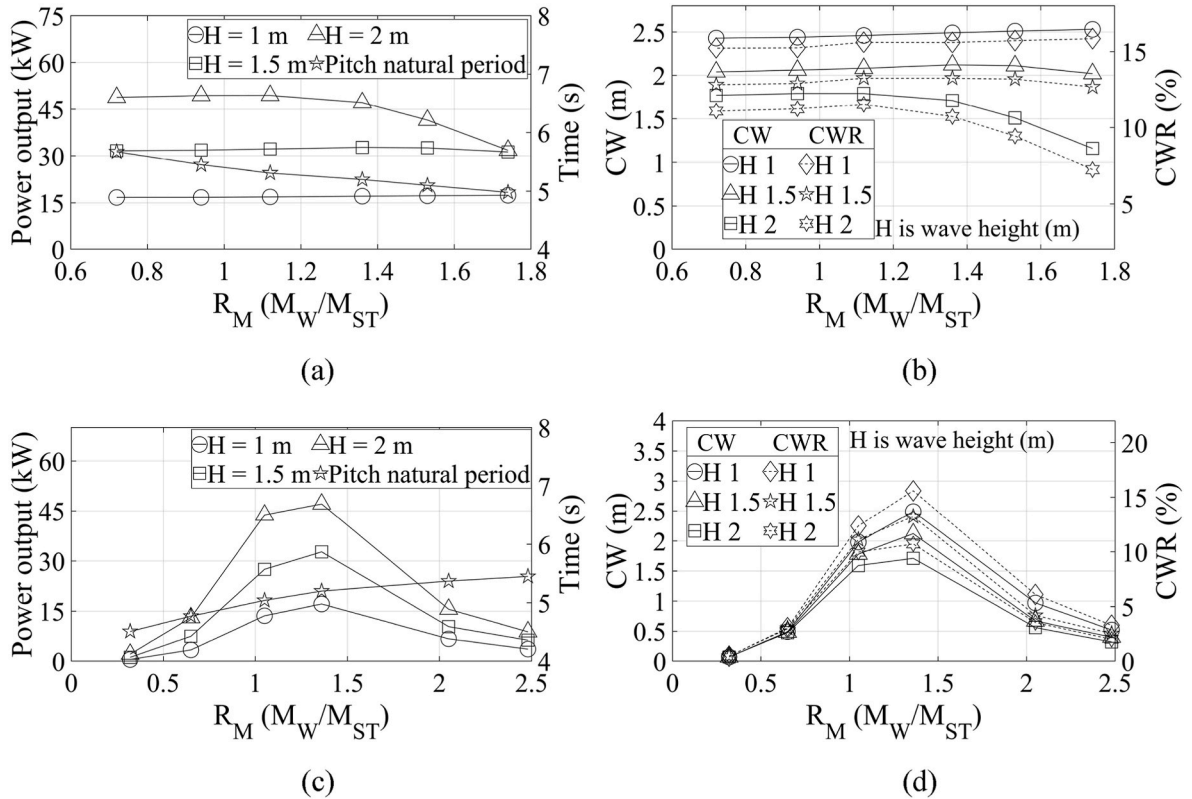


Fig. 12. With constant outer diameter (a) Power output versus R_M , (b) CW and CWR versus R_M ; with changing outer diameter of all tubes (c) Power output versus R_M , (d) CW and CWR versus R_M .

peaks at $R_M = 1.4$, which corresponds to the mass ratio of the base case structure.

Variation of the extracted power with respect to time period for various wave heights are plotted in Fig. 13a for the base case structure of Table 1. The available power in the waves depends on the wave height and its time period. Conforming this, the extracted power has been noted to increase with the wave height. Also, the peak of the energy extraction can be observed at 7 s, which is well expected as the water column has been tuned to that. In Fig. 13b, the plot of the CW and CWR is showing similar trend, i.e., the peak values appear corresponding to the tuning period of 7 s.

6. Extreme environmental condition check

The extreme environmental condition of the whole system has been checked for the extreme sea state presented in Table 3, considering both regular and irregular waves. For irregular waves, 10 random realizations, each of 3 h, have been simulated using the same JONSWAP spectrum. From each of these simulations, the maximum values (relative to the still conditions) of surge, heave, pitch and mooring line force variables have been obtained. The average of the 10 such maximum values has been taken as the expected maximum value of the variable. These results are shown in Table 6.

As mentioned earlier, the mean position of the structure will be shifted due to the drift force. The natural periods at the drifted condition are shown in Table 5. As expected, the irregular sea condition exhibit less mean drift and line tension compared to the regular waves. The maximum line tension can be noticed to remain within its allowable limit as presented in Table 2, and the minimum tension appears not to cause any slack condition in the lines. However, the maximum pitch motion in both regular and irregular sea states appear to be large for the chosen level of damping.

It can be noticed that the values of all parameters are relatively close for both the regular and irregular waves, except the line tension. This can be attributed to the mean drift position of the structure which is 13 m in regular waves and 3.15 m in irregular waves. This causes larger deformation in the mooring lines in regular waves, which consequently produce larger mean tension. The variation of the maximum pitch during the extreme event (regular wave) with different damping constants is shown in Table 7 keeping all other data same as in Table 1. It can be noticed that the minimum pitch appears to be 18.95° , and this corresponds to a fully restrained water column. The line tension is also minimum for this condition.

The time series of amplitudes (surge, heave, pitch and line tension) obtained from the extreme sea state simulation have been converted into frequency domain using Fourier Transformation after removing their mean value. The amplitude spectrum are plotted in semi-log scale in Fig. 14 for both regular (Fig. 14a–d) and random waves (Fig. 14e–h).

In the plots corresponding to the regular waves, peaks can be visible at the wave frequency (0.12 Hz) for the surge, heave, pitch, and line tension. Smaller peaks at its harmonics are also visible in these plots. Similar trend can be noticed in all the plots corresponding to the random

Table 6

Results of extreme environmental condition analysis.

Parameters	Values
With a regular wave ^a	
Maximum wave height, wave period	7.6 m, 8.3 s
Maximum surge motion (m)	16.10
Maximum heave motion (m)	2.42
Maximum pitch motion (m)	35.11°
Maximum water column motion (m)	7.52
Maximum line tension (kN)	184.94
With irregular wave (3-h duration) ^a	
Mean of maximum wave height, wave period	7.71 m, 8.26 s
Mean of maximum surge motion (m)	13.58
Mean of maximum heave motion (m)	3.54
Mean of maximum pitch motion (°)	31.12
Mean of maximum water column motion (m)	7.03
Maximum line tension (kN)	155.47

^a Refer Table 3 for the extreme sea state.

Table 7

Pitch amplitude at various damping under extreme (regular) wave condition.

Damping (kNs/m)	Maximum Pitch motion (°)	Maximum water column motion (m)	Maximum tension in moorings (kN)
17.5	35.11	7.52	184.94
50	23.26	3.95	116.45
80	20.68	2.68	100.83
No movement allowed	18.95	0.00	69.49

waves as well, i.e., peaks at 0.11 Hz are clearly visible, while the rise at the locations of the harmonics can be regarded to be small, i.e., large dynamic amplifications at the system natural frequencies are not present in the responses. The natural frequencies of heave (0.15 Hz) and pitch (0.16 Hz) are very close to natural frequency of water column (0.14 Hz). Since optimum damping corresponds to this frequency, the peaks at these frequencies become very small. Also, the responses in the frequency domain do not point out any other noticeable response due to the nonlinear interactions of various terms present in the mathematical model.

In the spectrum plots of the surge motion and line tension (Fig. 14 e and h), a peak at 0.015 Hz can also be noticed which corresponds to the surge natural frequency of the moored structure. This has been analysed further in Fig. 15 where the plot consists of the resultant surge in time domain, and spectrum converted into time domain after filtering out peaks for surge frequency and wave frequency separately. The resultant surge clearly shows displacements occurring due to the combination of drift motion and wave motion. Similar characteristics can also be noticed for the line tension plot.

7. Conclusion

In this paper, the application of a U-tube TLC based WEC unit stationed with a mooring system at 50 m water depth producing around

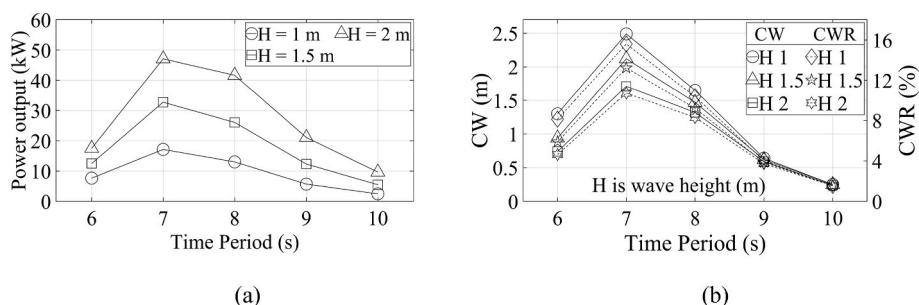
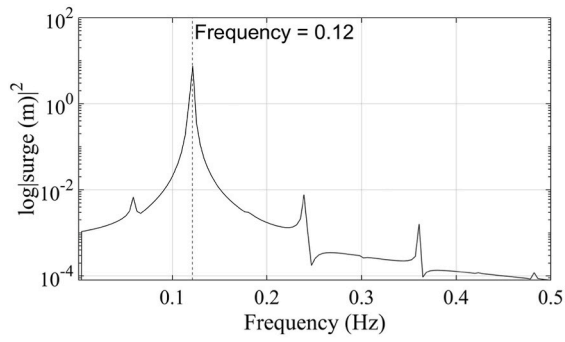
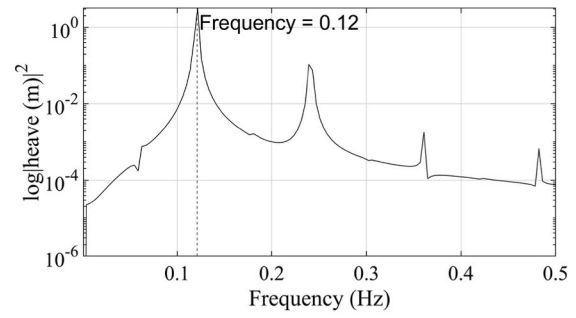


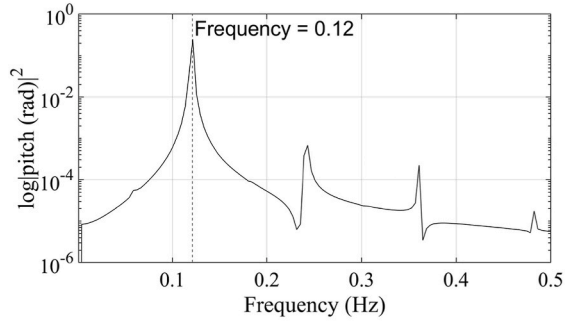
Fig. 13. For various wave heights (a) power output versus wave periods, (b) CW and CWR versus wave periods.



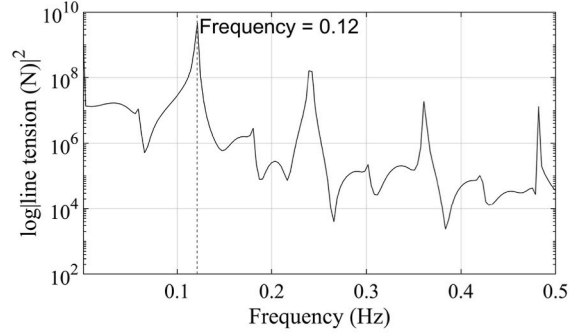
(a)



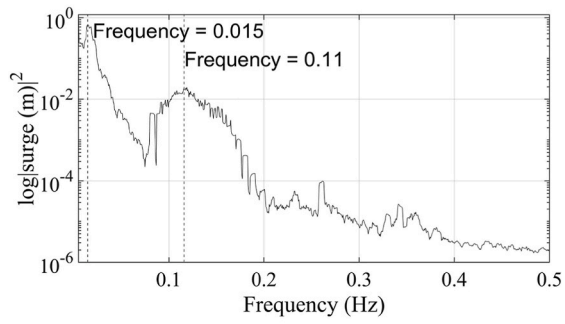
(b)



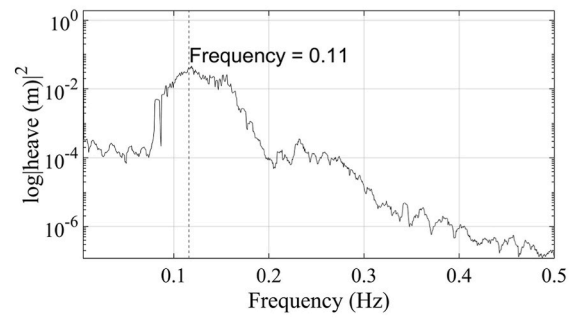
(c)



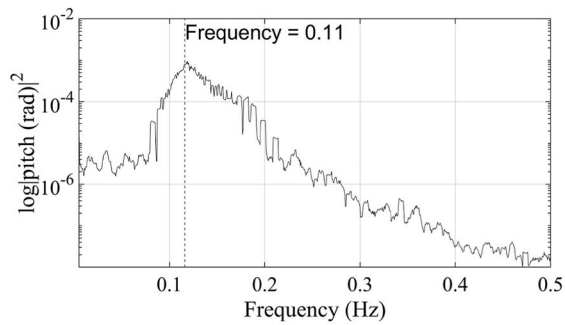
(d)



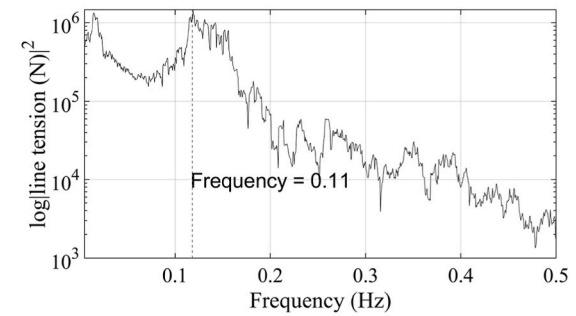
(e)



(f)



(g)



(h)

Fig. 14. Frequency domain response in extreme regular wave (a) surge, (b) heave, (c) pitch, (d) line tension; in extreme irregular wave (e) surge, (f) heave, (g) pitch, (h) line tension.

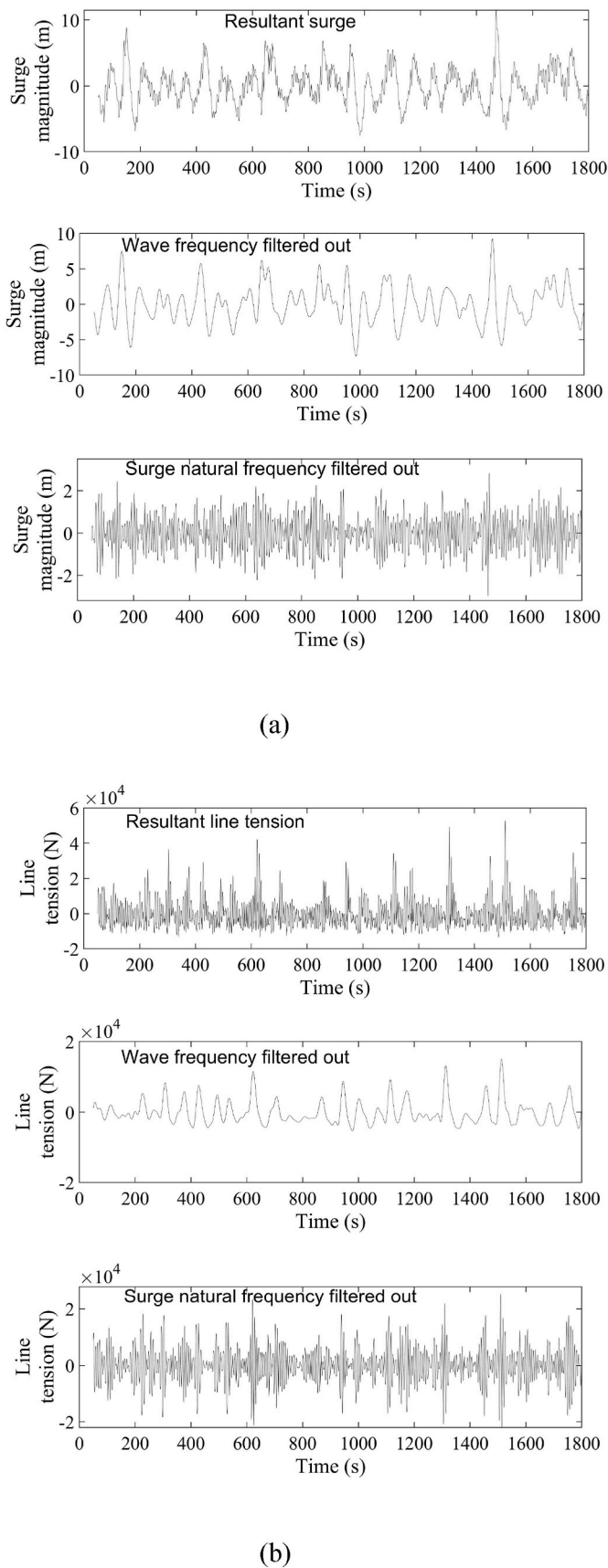


Fig. 15. Spectrum filtered out for surge and wave frequency separately and converted into time domain for (a) surge motion, (b) line tension.

50 kW power has been investigated. A coupled analysis approach has been adopted where the floating body (U-tube) has been modelled in a standard hydrodynamic tool, to which forces exerted by the motion of the water column has been added through a dll file. The water column has been tuned considering a typical sea state of 1–2 m wave height and 7 s wave period, which is commonly available at many coastal regions around the world. The PTO has been assumed to be a linear damper. Simpler 2 DOF liquid column damper models well established in the literature have been used to validate the numerical model of the whole system. The validated model has been used to study the sensitivity of the extracted power, capture width and the capture width ratio with respect to various design parameters defining the geometry of the U-tube structure.

It has been found that the maximum mooring line tension remains within its allowable limit during an extreme environmental event. Attaching clump weights to the mooring lines appears to be a good option to avoid slack in the mooring lines. Due to the asymmetry in the mooring system effected by the drifted mean position of the floating structure, the surge natural period has been changed, however the natural periods in heave and pitch are noted to remain unaffected.

The extracted power has been found to increase with the wave height as it induces more energy to the water column. However, the capture width and the capture width ratio reduce with increasing wave height, revealing that the system cannot capture more energy at higher excitation. This, however, requires further detailed investigations with different damping constants, tuning, and tube configurations.

It has been summarized that the energy capture can be improved by (i) choosing lower R_L values, (ii) by taking $R_A = 1$, and (iii) by considering the same R_M for which the damping constant delivers peak power output. However, these choices could be dominated by practical considerations as well.

The extreme condition has been studied using both regular and irregular waves. The extreme line tension is noticed to be larger for regular waves since the mean drift position is larger in regular waves.

The frequency domain plots clearly show peaks at the expected positions, which are the surge natural frequency, wave frequency and its harmonics. No other significant peaks have been noticed pointing out that the system behaviour is not dominated by the nonlinear terms present in its mathematical model. The peak at the surge natural frequency is visible only in the spectrum of surge and line tension obtained from irregular waves.

The current study does not look into the effect due to the changing environmental conditions in details. No optimization study on the extracted power has been attempted currently. No experimental verification of the current mathematical model or detailed investigation into any possible nonlinear behaviour just outside the domain of the operating condition is included. Also, no structural analysis of the tube for its bending due to the hydrodynamic loads has been included. The mathematical model developed in this study can be used to couple offshore wind turbines with TLCD for enhancing their stability.

CRediT authorship contribution statement

Gaurav Bhaskar: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **Arunjyoti Sarkar:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

Surge DOF :

$$\begin{aligned} & (L_A \rho + M) \ddot{x} + (-2 \rho A \sin(p) w + \rho A L_H \cos(p)) \ddot{w} + (-\rho A \sin(p) L_H w + (-\rho A w^2 \\ & + (-L_V^2 + 2 L_C L_V + L_H L_C) A \rho - M L_T + M L_C) \cos(p)) \ddot{p} + ((\rho A w^2 + (L_V^2 - 2 L_C L_V \\ & - L_H L_C) A \rho + M L_T - M L_C) \sin(p) - \rho A L_H \cos(p) w) \dot{p}^2 + (-2 \rho A \sin(p) L_H \\ & - 4 \rho A \cos(p) w) \dot{p} \dot{w} - 2 \rho A \sin(p) \dot{w}^2 + K_{surge} x = 0 \end{aligned}$$

W DOF :

$$\begin{aligned} & \rho A (L_H \cos(p) - 2 \sin(p) w) \ddot{x} + \rho A L \ddot{w} + \rho A (L_H \sin(p) + 2 \cos(p) w) \ddot{y} + \rho A L_H (L_C \\ & + L_V) \ddot{p} + \rho A w (2 L_C - 2 L_V) \dot{p}^2 + \rho g A L_H \sin(p) + 2 \rho g A \cos(p) w \end{aligned}$$

Heave DOF :

$$\begin{aligned} & (2 \rho A \cos(p) w + \rho A \sin(p) L_H) \ddot{w} + (L_A \rho + M) \ddot{y} + (\rho A L_H \cos(p) w + (-\rho A w^2 + \\ & - L_V^2 + 2 L_C L_V + L_H L_C) A \rho - M L_T + M L_C) \sin(p)) \ddot{p} + (-\rho A \sin(p) L_H w + (-\rho A w^2 \\ & + (-L_V^2 + 2 L_C L_V + L_H L_C) A \rho - M L_T + M L_C) \cos(p)) \dot{p}^2 + (2 \rho A L_H \cos(p) \\ & - 4 \rho A \sin(p) w) \dot{p} \dot{w} + 2 \rho A \cos(p) \dot{w}^2 + K_{heave} y + \rho L A g + M g = 0 \end{aligned}$$

Pitch DOF :

$$\begin{aligned} & (-\rho A \sin(p) L_H w + (-\rho A w^2 + (L L_C - L_V^2) A \rho - M L_T + M L_C) \cos(p)) \ddot{x} + \rho A L_H (L_C \\ & + L_V) \ddot{w} + ((-\rho A w^2 + (L L_C - L_V^2) A \rho - M L_T + M L_C) \sin(p) + \rho A L_H \cos(p) w) \ddot{y} \\ & + \left((2 L_V - 2 L_C) A \rho w^2 + \left(L L_C^2 - 2 L_C L_V^2 + \frac{L_H^2 L_V}{2} + \frac{L_H^3}{12} + \frac{2 L_V^3}{3} \right) A \rho + M L_T^2 \right. \\ & \left. + M L_C^2 + M L_S - 2 M L_T L_C \right) \ddot{p} + (4 L_V - 4 L_C) A \rho w \dot{p} \dot{w} + (-\rho g A w^2 + (L g L_C \\ & - g L_V^2) A \rho - M g L_T + M g L_C) \sin(p) + \rho g A w L_H \cos(p) + K_{pitch} p = 0 \end{aligned}$$

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